

Representational Measurement of Similarity: The Additive Difference Model, Revisited

Arthur Liske, Mathematik

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Abstract text.....

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1 Introduction

2 Proximity Structures and Metrics

3 Segmental Additivity

3.1 Segmentally Additive Metrics

3.2 Segmentally Additive Proximity Structures

4 Additive Difference Models and Structures

4.1 Additive Difference Model

4.2 Additive Difference Structure

5 Combining Segmental Additivity with the AD-Model

Theorem 5.1 (Necessary and sufficient conditions for additive difference metrics).

Suppose $\langle A, \succsim \rangle$ is an additive-difference proximity structure such that f is the additive-difference representation from ??, i.e.,

$$f(a, b) = \sum_{i=1}^n F_i(|\varphi_i(a) - \varphi_i(b)|).$$

Suppose further that the domain of each F_i is a real interval and $F_i(0) = 0$, $i = 1, \dots, n$ (through transformation, since F_i are interval scales).

For $\langle A, \succsim \rangle$ to be compatible with a metric, and additionally be additive on the coordinate axes,

1. it is necessary that there exists a continuous, convex function F , such that $F_i(x) = F(t_i x)$, $t_i > 0$, for all x in the domain of F_i , $i = 1, \dots, n$; and
2. sufficient that, in addition (to the conditions in 1.), the function $\log F(e^x)$ is convex.

Remark 5.2 (Additivity on the Coordinate Axes). Additivity on the coordinate axes means the following: If $a, b, c \in A$ differ only on one factor A_i , then the triangle inequality of the metric holds with equality. In mathematical terms:

$$a_j = b_j = c_j \text{ for all } j \neq i \text{ and } (a, c) \succsim (a, b), (b, c) \Rightarrow \delta(a, b) + \delta(b, c) = \delta(a, c)$$

In the following lemmas until the proof of this theorem is finished everything will be assumed as in theorem 5.1. **Todo: checken bis wohin genau**

Corollary 5.3 (Form von δ).

Seien die Voraussetzungen wie in Satz ??. Dann wird die Metrik δ dargestellt durch

$$\delta(a, b) = F^{-1} \left\{ \sum_{i=1}^n F(|\varphi_i(a_i) - \varphi_i(b_i)|) \right\} \quad (1)$$

wobei F stetig, konvex und steigend ist.

Proof of theorem 5.1 1. According to the assumptions of the theorem, there exists a metric δ on A and an increasing function G , such that for all $a, b \in A$,

$$\delta(a, b) = G \left(\sum_{i=1}^n F_i(|\varphi_i(a_i) - \varphi_i(b_i)|) \right).$$

Assume a , b , and c differ only in the i -th factor and $a|b|c$. Thus, we can use additivity on the i -th coordinate axis for a, b, c which yields

$$G(F_i(|\varphi_i(a_i) - \varphi_i(b_i)|)) + G(F_i(|\varphi_i(b_i) - \varphi_i(c_i)|)) = G(F_i(|\varphi_i(a_i) - \varphi_i(c_i)|)).$$

Defining $|\varphi_i(a_i) - \varphi_i(b_i)| =: x$ and $|\varphi_i(b_i) - \varphi_i(c_i)| =: y$ yields

$$G[F_i(x)] + G[F_i(y)] = G[F_i(x + y)].$$

Further, define $H_i(x) = G[F_i(x)]$. This reduces the above equation to $H_i(x) + H_i(y) = H_i(x + y)$ for all x, y in the domain of F_i , $1 \leq i \leq n$. This means, that H_i must be linear and since the values are positive, the only monotonic solution to this equation is $H_i(x) = t_i x$, $t_i > 0$. Since G is increasing, it has an inverse G^{-1} satisfying

$$G^{-1}[H_i(x)] = G^{-1}[t_i x] = F_i(x)$$

where F_i is defined. Since G^{-1} is independent of i , the F_i can only differ in range and in the unit of the domain (i.e. the t_i which adjusts the domain where G^{-1} is defined). Without loss of generality, we can assume that the range of F_n is the largest. Since the range of F_n includes the range of F_i , we can set $F = F_n$ and obtain $F_i(x) = F_n(t_i x) = F(t_i x)$ for $t_i > 0$ and for every x in the domain of F_i . Continuity and convexity will be proven in the next lemmas. $\square$

Todo: sagen dass conditions wie oben sind

Lemma 5.4.

$$F(\alpha + \beta + \gamma) + F(\gamma) \geq F(\alpha + \gamma) + F(\beta + \gamma).$$

Proof. In the triangle inequality,

$$F^{-1} \left[\sum_{i=1}^n F(\alpha_i) \right] + F^{-1} \left[\sum_{i=1}^n F(\beta_i) \right] \geq F^{-1} \left[\sum_{i=1}^n F(\alpha_i + \beta_i) \right],$$

let $\alpha_1 = \gamma$, $\alpha_2 = F^{-1}(F(\alpha + \gamma) - F(\gamma))$, $\alpha_j = 0$ for $3 \leq j \leq n$ and $\beta_1 = \beta$, $\beta_j = 0$ for $2 \leq j \leq n$. Hence,

$$\begin{aligned} F(\alpha + \beta + \gamma) &= F \left[F^{-1} \left[\sum_{i=1}^n F(\alpha_i) \right] + F^{-1} \left[\sum_{i=1}^n F(\beta_i) \right] \right] \\ &\geq \sum_{i=1}^n F(\alpha_i + \beta_i) \quad (\text{by the triangle inequality}) \\ &= F(\beta + \gamma) + F(\alpha + \gamma) - F(\gamma). \end{aligned}$$

$\square$

Lemma 5.5.

F is continuous.

Proof. Since F is increasing, all discontinuities are gaps. Real analysis yields, that real monotonic functions have at most countable many gaps. We will derive a contradiction from this.

Assume $F(\alpha)$ is the lower boundary of a gap, then there exists $\epsilon > 0$ such that for all $\delta > 0$, $F(\alpha + \delta) - F(\alpha) > \epsilon$. However, by lemma 5.4, for every $\beta > 0$,

$$F(\alpha + \beta + \delta) - F(\alpha + \beta) \geq F(\alpha + \delta) - F(\alpha) > \epsilon,$$

and so there is a gap at any $\alpha + \beta$, and F would have to be discontinuous everywhere. This is impossible, so F must be continuous. $\square$

Lemma 5.6.

Suppose F is a strictly increasing function defined on the nonnegative reals. Then F is convex iff for all $\alpha, \beta, \gamma \geq 0$ the inequality from 5.4 holds, i.e.,

$$F(\alpha + \beta + \gamma) + F(\gamma) \geq F(\alpha + \gamma) + F(\beta + \gamma).$$

Proof.

Part 1 ($\Rightarrow$).

Suppose, F is convex, and let $p = \alpha/(\alpha + \beta)$, then

$$\begin{aligned} p(\alpha + \beta + \gamma) + (1 - p)\gamma &= \alpha + \gamma \\ (1 - p)(\alpha + \beta + \gamma) + p\gamma &= \beta + \gamma. \end{aligned}$$

Hence, by convexity,

$$\begin{aligned} F(\alpha + \gamma) + F(\beta + \gamma) &= F[p(\alpha + \beta + \gamma) + (1 - p)\gamma] + F[(1 - p)(\alpha + \beta + \gamma) + p\gamma] \\ &\leq pF(\alpha + \beta + \gamma) + (1 - p)F(\gamma) + (1 - p)F(\alpha + \beta + \gamma) + pF(\gamma) \\ &= F(\alpha + \beta + \gamma) + F(\gamma). \end{aligned}$$

Part 2 ($\Leftarrow$).

To prove the converse, we will use the continuity of F by lemma 5.5. Suppose, $\alpha > \beta$, then

$$\begin{aligned} F(\alpha) + F(\beta) &= F\left(\frac{\alpha - \beta}{2} + \frac{\alpha - \beta}{2} + \beta\right) + F(\beta) \\ &\geq 2F\left(\frac{\alpha - \beta}{2} + \beta\right) && \text{(by lemma 5.5)} \\ &= 2F\left(\frac{\alpha + \beta}{2}\right), \end{aligned}$$

which is equivalent to convexity for a continuous F . $\square$

5.0.1 Convexity of δ Balls

6 Discussion

References

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